

6(a)(i)	thylakoid (membranes) / lamella(e) / grana ;	1
6(a)(ii)	<i>any three from:</i> 1 ref. to LHC / light harvesting complex ; 2 (named) accessory pigments ; 3 pass, light / photons, to, reaction centre / chlorophyll <u>a</u> / primary pigment/chlorophyll P680 ; 4 <u>non-cyclic</u> photophosphorylation ; 5 more / different, wavelengths / energy frequencies, absorbed / used ;	3
6(b)	<i>any two from:</i> 1 as (concentration of) cadmium (ions) increases PSII <u>activity</u> decreases ; 2 supporting figures comparative quote with units ; 3 concentration rises by, order of magnitude / factor of 10, each time ;	2
6(c)(i)	<i>any four from:</i> D1 in month A functional chlorophyll concentration is higher in unpolluted lake (than polluted lake) ; D2 in month B functional chlorophyll concentration is same in both (lakes) ; D3 in both (lakes) functional chlorophyll concentration is higher in month A than in month B ; E4 in polluted lake copper ions, inhibit / damage / disrupt, chlorophyll ; E5 in month B, protocists are, dormant / spores / resist entry of ions / AW ; E6 month A =, summer / <u>hotter</u> / <u>higher</u> light intensity / <u>longer</u> days ; 7 AVP ; ref. flooding decreases Cu^{2+} concentration in month B evaporation increases Cu^{2+} concentration in month A	4
6(c)(ii)	substitute for, magnesium ion / Mg^{2+} ;	1
6(d)	<i>any four from:</i> 1 spot / extract, placed on, pencil line / base line / line of origin ; 2 repeat / concentrate, spot / extract; 3 end / base, of chromatogram, suspended / placed, in solvent ; 4 R_f value = distance moved by, spot / pigment / solute ÷ distance moved by solvent (front) ; 5 compare with known R_f value to identify, chlorophyll a / pigment ; 6 detail of method ; e.g. cover to stop evaporation of solvent remove chromatogram before solvent front reaches top	4